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GitHub Link for all code:

https://github.com/Kevin-Maniar/06Aug_Kevin_SE/tree/main/C%20_%20Assignment%20_%20Code

Module – 2 | Session – 1

Q – 1

Write a short paragraph in your own words defining what a 'program' is and give one real-life example from any app you use daily (like Instagram, Zomato, or Paytm).

Program: -

- ⇨ Program is a set of instructions written for a computer or mobile device to perform a specific task. Program tells a device what to do and how to do it.
- ⇨ For ex. Paytm – Allow user to make payments, receive payments, pay bills, and check for balance.

Q – 2

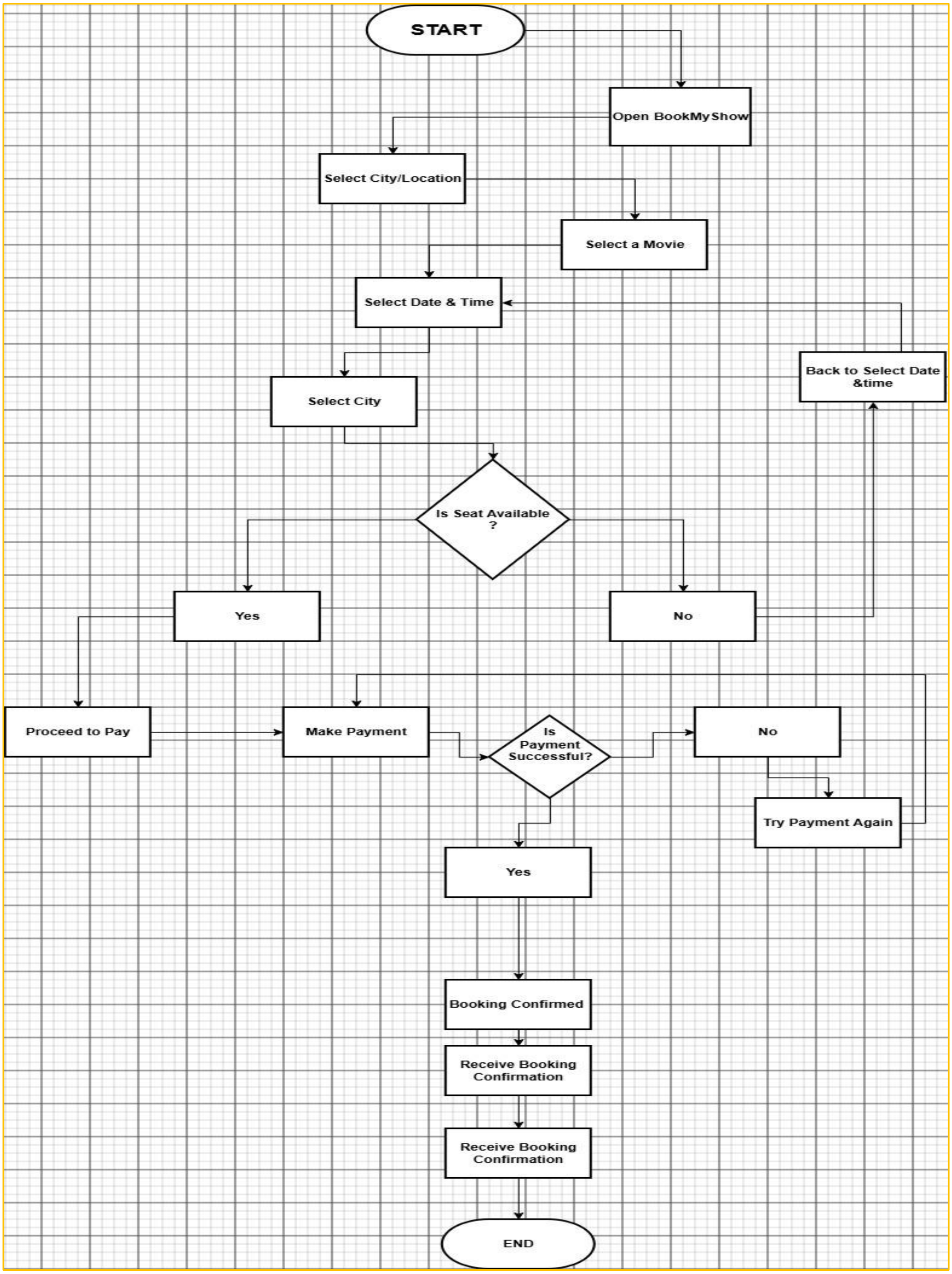
List three Programming languages you have heard of, and for each mention whether it is usually compiled, interpreted, or can be both.

Programming Languages: Java, Python, JavaScript, C++

- ⇨ Java: Compiled
- ⇨ Python: Interpreted
- ⇨ JavaScript: Both Compiled and Interpreted
- ⇨ C++: Compiled

Q – 3

Draw a flowchart (on paper or using any online tool) for the process of booking a movie ticket on BookMyShow, starting from opening the app to receiving the booking confirmation.


START
Open BookMyShow
Select City/Location
Select a Movie
Select Date & Time
Back to Select Date &time
Select City
Is Seat Available ?
Yes
No
Proceed to Pay
Make Payment
Is Payment Successful?
No
Try Payment Again
Yes
Booking Confirmed
Receive Booking Confirmation
Receive Booking Confirmation
END

Q – 4

Write down the steps (as an algorithm, using simple numbered instructions) for ordering food on Swiggy — from opening the app to completing payment.

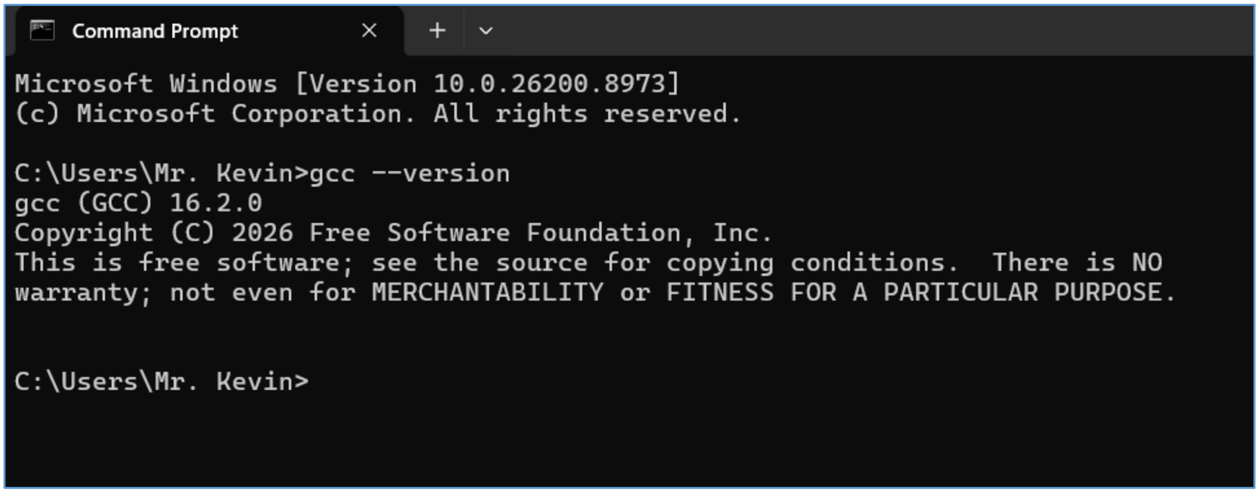
⇨ Algorithm: Ordering food from swiggy

1. Start
2. Open the Swiggy App
3. Log in or Signup
4. Search for the Restaurant or select from the list
5. Select the food items you want to order
6. Add the selected items into the cart
7. Open the cart
8. Enter delivery location and confirm
9. Place order
10. Select the Payment method
11. Complete payment
12. If Payment is Successful, order is confirmed
13. Receive the order confirmation
14. End

Session – 2

Q – 1

Install GCC on your computer and verify the installation by running gcc --version in your terminal or command prompt.



Q – 2

Open VS Code or DevC++ and create a new file named hello.c. Write a C program that prints 'Hello, Instagram World!' to the console.

```
E:\06Aug_Kevin_SE\C _ Assign

Hello World!
--------------------------------
Process exited after 0.131 seconds with return value 0
Press any key to continue . . .
```

Q – 3

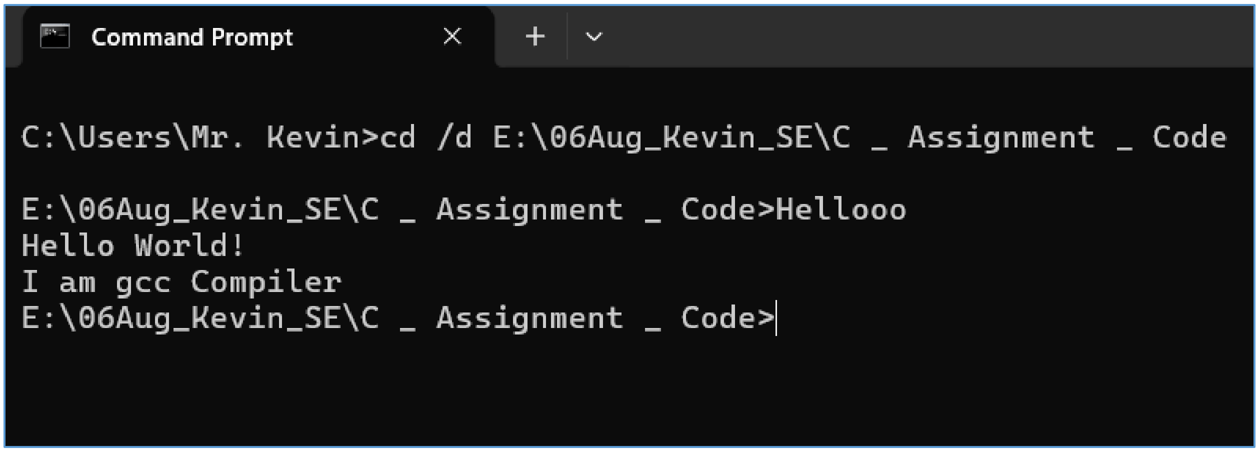
Identify and label the header, main function, and return statement in your hello.c program using comments.

Hellooo.c

```c
#include<stdio.h> // header file
int main() // main function where program start
{
    printf("Hello World!");
    return 0; // return statement
}
```

Q – 4

Compile your hello.c file using the GCC command line (gcc hello.c -o hello) and run the output file to display the message in your terminal.



Session: 3

Q – 1 Declare variables for a Flipkart product: productName (as a string), price (float), and rating (double). Assign sample values and print each variable with its data type.

S3T1_FP.c

```c
#include <stdio.h>

#define get_type(X) _Generic((X), \
    float: "float", \
    double: "double", \
    char*: "string ", \
    default: "unknown type")

int main()
{
    char productName[] = "Samsung Galaxy S24";
    float price = 59999.99;
    double rating = 4.5;

    printf("Product Name: %s\n", productName);
    printf("Data Type: %s\n\n", get_type(productName));

    printf("Price: %.2f\n", price);
    printf("Data Type: %s\n\n", get_type(price));

    printf("Rating: %.1lf\n", rating);
    printf("Data Type: %s\n", get_type(rating));

    return 0;
}
```

E:\06Aug_Kevin_SE\C _ Assign

```
Product Name: Samsung Galaxy S24
Data Type: string

Price: 59999.99
Data Type: float

Rating: 4.5
Data Type: double

--------------------------------
Process exited after 0.06464 seconds with return value 0
Press any key to continue . . .
```

Q – 2

Create a constant variable to store the GST rate (for example, 18%) and use it to calculate the final price of a Zomato order with a given base price.

Constraint: The GST rate must not be changeable after its initial assignment.

S3T1_FP.c S3T2_Const.cpp

```c
#include<stdio.h>
int main ()
{
    const float GST_RATE = 0.18; //Gst rate 18%
    float base_price ;

    printf("Enter Base price of food:");
    scanf("%f",&base_price);

    double gst = GST_RATE * base_price ;
    float final_price = base_price + gst;

    printf("final price is :%.2f",final_price);
    return 0;
}
```

E:\06Aug_Kevin_SE\C _ Assign

```
Enter Base price of food:1200
final price is :1416.00
--------------------------------
Process exited after 5.663 seconds with return value 0
Press any key to continue . . .
```

Q – 3

Write a program that stores your favorite Spotify playlist's name (string), total number of songs (int), and average song duration in minutes (float). Print all values in a single formatted sentence.

S3T1_FP.c S3T2_Const.cpp S3T3_Playlist.cpp

```c
/*Write a program that
stores your favorite Spotify playlist's name (string),
total number of songs (int),
and average song duration in minutes (float).
Print all values in a single formatted sentence.*/

#include<stdio.h>
int main()
{
    char playlist[10];
    int total_songs;
    float avg_song_duration;

    printf("Enter Your fav Spotify playlist's Name:");
    scanf("%s",&playlist);
    printf("total_songs are :");
    scanf("%d",&total_songs);
    printf("Avg Duration:");
    scanf("%f",&avg_song_duration);

    printf("Playlist Name: %s\n",playlist);
    printf("Total Songs : %d\n",total_songs);
    printf("Total Duration: %.0f Minutes\n",avg_song_duration);

    return 0;

}
```

E:\06Aug_Kevin_SE\C _ Assign

```
Enter Your fav Spotify playlist's Name:Chill_Vibes
total_songs are :20
Avg Duration:4
Playlist Name: Chill_Vibes
Total Songs : 20
Total Duration: 4 Minutes

--------------------------------
Process exited after 17.47 seconds with return value 0
Press any key to continue . . .
```

Q – 4

Identify and correct the invalid variable names from this list:

1stPlayer, player_age, $score, total-marks, userName.

- ⇨ Invalid Variable names are : 1stPlayer , $score , total-marks
- ⇨ Correct Version : first_player , score , total_marks

Session: 4

Que – 1

Create a simple JavaScript function called calculateTotal that takes two numbers: itemPrice and quantity and returns the total bill amount using arithmetic operators.

S3T3_Playlist.cpp S4T1_Arithmatic_operator.cpp

```
/*Create a simple JavaScript function called
calculateTotal that takes two numbers:
itemPrice and quantity, and returns the
total bill amount using arithmetic operators.*/
#include<stdio.h>
int main()
{


    float itemPrice;
    int Quantity;
    float Cal_Total;

    printf("Enter Item Price:");
    scanf("%f",&itemPrice);
    printf("Enter Quantity:");
    scanf("%d",&Quantity);

    Cal_Total = itemPrice * Quantity;
    printf("Total Bill Amount is: %.2f INR",Cal_Total);

    return 0;
}
```

E:\06Aug_Kevin_SE\C _ Assign

```
Enter Item Price:120
Enter Quantity:3
Total Bill Amount is: 360.00 INR
--------------------------------
Process exited after 2.473 seconds with return value 0
Press any key to continue . . .
```

Q – 2 Build a Flipkart-style discount calculator given product price, discount percentage, and a boolean isMember, use arithmetic and logical operators to calculate the final price (apply an extra 5% off if isMember is true).

S4T2_Discount_Calculator.cpp

```
#include<stdio.h>
int main()
{
    /*Build a Flipkart-style discount calculator:
    given product price,
    discount percentage,
    and a boolean isMember,
    use arithmetic and logical
    operators to calculate the final price
    (apply an extra 5% off if isMember is true). */

    float product_price, discount_percentage;
    float final_price, discount_price;
    int isMember;

    printf("Enter Product Price:");
    scanf("%f",&product_price);

    printf("Enter Discount Percentage:");
    scanf("%f",&discount_percentage);

    printf("Are You Member: (1 for Yes , 0 for NO)");
    scanf("%d",&isMember);

    // Count Normal price
     discount_price = product_price * (discount_percentage/100);
    //Price After Discount
    final_price = product_price - discount_price;
    //Extra 5% discount

    if (isMember == 1)
    {
        final_price = final_price -(final_price *5/100);
        printf("Your Final price is :%.2f",final_price);
    }
    else
    {
        printf("Final Price = %.2f$",final_price);
    }

}
```

```
E:\06Aug_Kevin_SE\C _ Assign

Enter Product Price:50000
Enter Discount Percentage:20
Are You Member: (1 for Yes , 0 for NO)1
Your Final price is :38000.00
--------------------------------
Process exited after 10.04 seconds with return value 0
Press any key to continue . . .
```

Q – 3

Write a function isEligibleForOffer that takes a user's age and total order value and returns true if the user is 18 or older AND the order value is above 500, otherwise false.

Hint: Use relational and logical operators together.

S4T2_Discount_Calculator.cpp S4T3_Relational_logical_oper.cpp

```
    /*Write a function isEligibleForOffer
    that takes a user's age and total order value,
    and returns true if the user is 18 or older AND
    the order value is above 500,
    otherwise false.*/

#include<stdio.h>

bool isEligibleForOffer (int age,float order_value)
{
    if(age>=18  && order_value>500)
    {
        return true;
    }
    else{
        return false;
    }
}
int main()
{
    int age;float order_value;
    printf("Enter Your Age:-");
    scanf("%d",&age);

    printf("Enter Your Total Order Value:-");
    scanf("%f",&order_value);

    if(isEligibleForOffer(age,order_value))
    {
        printf("YOU ARE ELIGIBLE FOR OFFER");
    }
    else{
        printf("YOU ARE NOT ELIGIBLE FOR OFFER!");
    }
    return 0;
}
```

```
E:\06Aug_Kevin_SE\C _ Assign

Enter Your Age:-18
Enter Your Total Order Value:-5000
YOU ARE ELIGIBLE FOR OFFER
--------------------------------
Process exited after 5.618 seconds with return value 0
Press any key to continue . . .
```

Q – 4

Given three variables: likes, comments, and shares (all numbers), write code to check if a post is 'trending' on Instagram (at least 1000 likes OR more than 200 comments AND at least 50 shares). Print the result.

S4T4_.cpp

```c
    /* Given three variables: likes, comments, and shares (all numbers),
    write code to check if a post is 'trending' on
    Instagram (at least 1000 likes OR more than
    200 comments AND at least 50 shares).
    Print the result. */

#include<stdio.h>
main ()
{
    int likes,comments,share;
    printf("Enter Total Number of Likes:");
    scanf("%d",&likes);
    printf("Enter Total Number of Comments:");
    scanf("%d",&comments);
    printf("Enter Total Number Share:");
    scanf("%d",&share);
    printf("-----------\n");
    printf("Liked By %d Fan \ncommented %d Fan \nShare By %d Fan\n",likes,comments,share);

    if(likes>=1000 || comments>200 && share>=50)
    {
        printf("-----------\n");
        printf("Result: Your Post is Trending! Congratulations");
    }
    else
    {
        printf("-----------\n");
        printf("Result: Your post is not Trending!!!");
    }
}
```

E:\06Aug_Kevin_SE\C _ Assign

```
Enter Total Number of Likes:300
Enter Total Number of Comments:400
Enter Total Number Share:51
-----------
Liked By 300 Fan
commented 400 Fan
Share By 51 Fan
-----------
Result: Your Post is Trending! Congratulations
--------------------------------
Process exited after 29.73 seconds with return value 0
Press any key to continue . . .
```

Q – 5

Write a code snippet that demonstrates the difference between pre-increment (++count) and post-increment (count++) by logging the values before and after using both on a followerCount variable.

S4T4_.cpp S4T5_.cpp

```c
    /*  Write a code snippet that
    demonstrates the difference between pre-increment
    (++count) and post-increment (count++)
    by logging the values before and after using both on a
    followerCount variable. */
#include<stdio.h>
int main()
{
    int follower = 100;

    printf("Pre Increment Follower ---> %d\n",++follower);
    printf("Pre Increment Follower ---> %d\n",++follower);
    printf("Pre Increment Follower ---> %d\n",++follower);

    printf("Post Increment Follower ---> %d\n",follower++);
    printf("Post Increment Follower ---> %d\n",follower++);
    printf("Post Increment Follower ---> %d\n",follower++);

    /* follower++ takes 100 first and then add +1
    ++follower add +1 in exist follower number

    pre increment give result by +1 and
    post increment give result by using old value first and then +1*/
    return 0;
}
```

E:\06Aug_Kevin_SE\C _ Assign

```
Post Increment Follower ---> 100
Post Increment Follower ---> 101
Post Increment Follower ---> 102
Pre Increment Follower ---> 104
Pre Increment Follower ---> 105
Pre Increment Follower ---> 106

--------------------------------
Process exited after 0.444 seconds with return value 0
Press any key to continue . . .
```

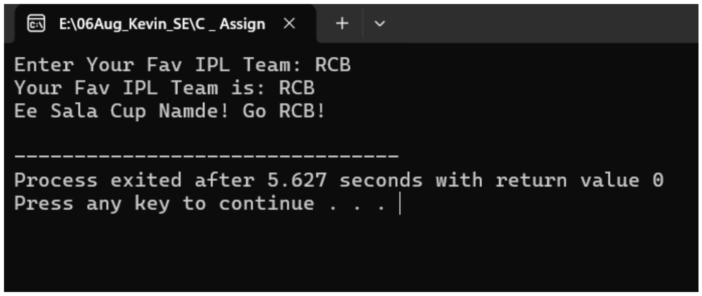
Session – 5

Que – 1

Create a simple IPL Fan Bot that takes your favorite IPL team name as input and uses if-else-if statements to print a unique cheer message for each team (e.g., 'Go Mumbai Indians!', 'Chennai Super Kings for the win!'). If the team is not recognized, print 'Team not found!'

```
#include <stdio.h>
#include <string.h>

int main() {

    char IPL_team[50];

    printf("Enter Your Fav IPL Team: ");
    scanf("%s", IPL_team);
    printf("Your Fav IPL Team is: %s\n", IPL_team);

    if (strcmp(IPL_team, "MI"    ) == 0) {
        printf("Go Mumbai Indians!\n");
    }
    else if (strcmp(IPL_team ,"GT") == 0) {
        printf("Aava De!\n");
    }
    else if (strcmp(IPL_team, "CSK") == 0) {
        printf("Chennai Super Kings for the win!\n");
    }
    else if (strcmp(IPL_team, "RCB") == 0) {
        printf("Ee Sala Cup Namde! Go RCB!\n");
    }
    else if (strcmp(IPL_team, "KKR") == 0) {
        printf("Korbo Lorbo Jeetbo Re! Go KKR!\n");
    }
    else {
        printf("Team Not Found!\n");
    }
    return 0;
}
```

Output:


E:\06Aug_Kevin_SE\C _ Assign
Enter Your Fav IPL Team: RCB
Your Fav IPL Team is: RCB
Ee Sala Cup Namde! Go RCB!

Process exited after 5.627 seconds with return value 0
Press any key to continue . . .

Que – 2

Build a Zomato-style food suggestion tool: take the user's preferred meal time ('breakfast', 'lunch', 'dinner', or 'snack') and use a switch-case statement to suggest a popular dish for that time. If the input doesn't match any meal, suggest 'Try some fruits!'

```
#include <stdio.h>
int main()
{
    int choice;
    printf("--- Food Suggestion Tool ---\n");
    printf("-----------------------------\n");
    printf("1..BreakFast\n2..Lunch\n3..Dinner\n4..Snack\n");
    printf("\n");
    printf("Enter Your Choice:");
    scanf("%d",&choice);
    printf("\n");

    switch (choice)
    {
        case 1:
            printf("Oh! BreakFast Time\n");
            printf("You can have\n");
            printf("1...OatMeal\n");
            printf("2...Eggs\n");
            printf("3...Poha\n");
            printf("4...Avocado\n");
            printf("5...Toast\n");
            printf("6...Milk\n");
        break;
        case 2:
            printf("Oh! Lunch Time\n");
            printf("You can have\n");
            printf("1...Roti With Dal\n");
            printf("2...Punjabi Dish\n");
            printf("3...Gujarati Thali\n");
            printf("4...Paneer Sabji\n");
        break;
        case 3:
            printf("Oh! Dinner Time\n");
            printf("You can have\n");
            printf("1..Vegetable Khichadi\n");
            printf("2..Snadwitch\n");
```

```c
            printf("3..Chhole Bhature\n");
            printf("4..Masala Dosa\n");
            printf("5..Idli\n");
        break;
        case 4:
            printf("Oh! Snack Time\n");
            printf("You can have\n");
            printf("1..Mixed Nuts & Almonds\n");
            printf("2..Roasted Makhna\n");
            printf("3..SoyaChunks\n");
            printf("4..Wafors\n");
            printf("5..Sev Mummra\n");
        break;
        default:
            printf("Have Some Fruits\n");
        break;
    }
    return 0;
}
```

Output: -

```
E:\06Aug_Kevin_SE\C _ Assign

Enter Your Fav IPL Team: RCB
Your Fav IPL Team is: RCB
Ee Sala Cup Namde! Go RCB!

--------------------------------
Process exited after 5.627 seconds with return value 0
Press any key to continue . . .
```

Que – 3

Create a Flipkart discount calculator that asks the user for the total cart amount. Use nested if statements to check: if amount > 2000, apply 20% discount; else if amount > 1000, apply 10% discount; else, no discount. Print the final amount to pay.

Hint: Use nested ifs to check each discount slab.

```
#include<stdio.h>
int main()
{
    float cart_amount;
    float final_price;
    printf("Enter the total cart amount:-");
    scanf("%f",&cart_amount);

    if(cart_amount>2000)
    {
        final_price = cart_amount*0.80;
        printf("You got 20%% Discount\n");
        printf("Your Final Amount is: %.2f\n",final_price);
    }
    else
    {
        if(cart_amount>1000)
        {
            final_price = cart_amount*0.90;
            printf("You got 10%% Discount\n");
            printf("Your Final Amount is: %.2f",final_price);
        }
        else{
            printf("No Discount\n");
            printf("Your Final Amount is: %2.f",cart_amount);
        }
    }
    return 0;
}
```

OutPut: -

```
E:\06Aug_Kevin_SE\C _ Assign

Enter the total cart amount:-3000
You got 20% Discount
Your Final Amount is: 2400.00

--------------------------------
Process exited after 3.774 seconds with return value 0
Press any key to continue . . .
```

Que – 4

Write a program that takes a user's age and checks eligibility for three things using if-else statements: if age >= 18, print 'Eligible for Driving License'; if age >= 21, also print 'Eligible for Credit Card'; if age >= 25, also print 'Eligible for Car Rental'. Print all applicable messages for the given age.

```
#include<stdio.h>
int main()
{
    int age;
    printf("Age:");
    scanf("%d",&age);

    if (age>=18)
        {
            printf("Lic\n");

    if(age>=21)
        {
            printf("credit\n");

    if(age>=25)
        {
            printf("Car\n");
        }
        }
        }
    else
    {
        printf("not Applicable");
    }
    return 0 ;
}
```

Output: -

```
E:\06Aug_Kevin_SE\C _ Assign
Default: E:\06Aug_Kevin_SE\C _ Assignment _ Code
\Session - 5\S5T4.exe
ctrl+alt+1
Eligible For Credit Card
Eligible For Car Rental
--------------------------------
Process exited after 5.607 seconds with return value 0
Press any key to continue . . .
```

```
E:\06Aug_Kevin_SE\C _ Assign
Enter Your Age:_19
Eligible For Driving Licence

--------------------------------
Process exited after 1.923 seconds with return value 0
Press any key to continue . . .
```

```
E:\06Aug_Kevin_SE\C _ Assign
Enter Your Age:_21
Eligible For Driving Licence
Eligible For Credit Card

--------------------------------
Process exited after 2.384 seconds with return value 0
Press any key to continue . . .
```

Session – 6

Que – 1

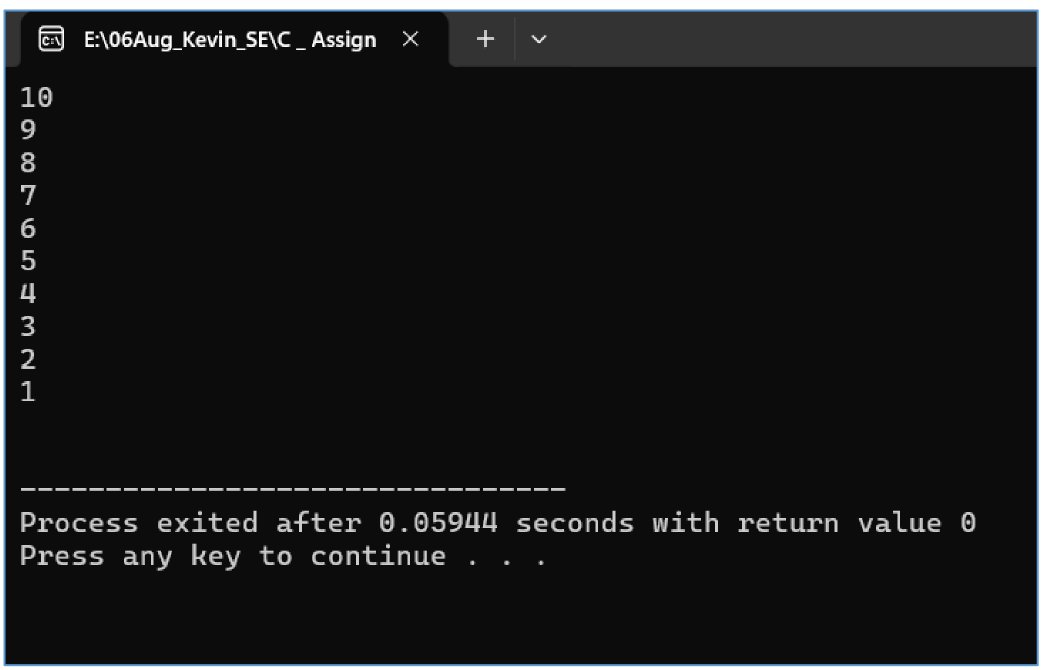
Write a program that prints the numbers from

10 down to 1 using a for loop,

similar to a countdown timer.

```c
#include<stdio.h>
int main()
{
    int i;
    for (i=10;i>=1;i--)
    {
        printf("%d\n",i);
    }
    printf("\n");
    return 0;
}
```

Output:



Que – 2

Create a menu-driven console app that lets the user:

1) View your favorite 3 IPL teams,

2) Add a new team, 3) Exit.

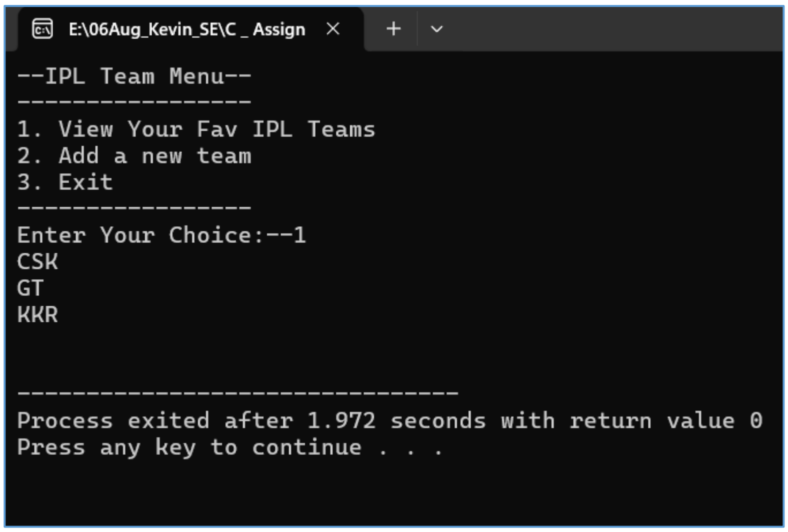
Use a while loop to keep showing the menu until the user chooses Exit.

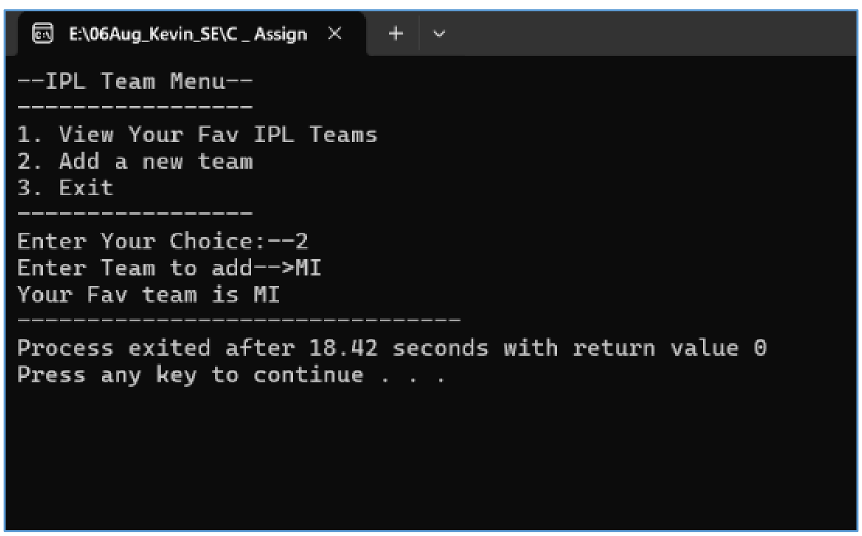
Hint: Use input() (or Scanner in Java) to get the user's choice each time.

```
#include<stdio.h>
int main()
{
    int choice;
    char team[15];
    while(1)
    {

    printf("--IPL Team Menu--\n");
    printf("-----------------\n");
    printf("1. View Your Fav IPL Teams\n");
    printf("2. Add a new team\n");
    printf("3. Exit\n");
    printf("-----------------\n");
    printf("Enter Your CHoice:--");
    scanf("%d",&choice);

    switch (choice)
    {
        case 1:

        //  printf("Your Favorite IPL Teams are as below\n");
            printf("CSK\n");
            printf("GT\n");
            printf("KKR\n");
            printf("\n");
            break;
        case 2:
            printf("Enter Team to add-->");
            scanf("%s",team);
            printf("Your Fav team is %s",team);
            break;
        case 3:
```

```
            printf("Exited");
            break;

    }
        return 0;
    }
}
```

Output →





Que – 3

Build a 'Guess the Song' game like Spotify — the program randomly picks a song name from a list and asks the user to guess it.

Use a do-while loop so the user can keep guessing until they get it right.

Constraint: Use at least 3 song names of your choice.

```
#include<stdio.h>
    #include<string.h>
    #include<stdlib.h>
    #include<time.h>

    int main ()
    {
        char song1[50] = "Sunn";
        char song2[50] = "ShapeOfYou";
        char song3[50] = "Levitating";
        char guess[50];
        char random_pick[50];

        int choice;
        srand(time(0));
        choice = rand()%3 ;

        switch(choice)
        {
            case 0:
                strcpy(random_pick,song1);
                printf("Hint: Singer is Dino James\n");
                break;
            case 1:
                strcpy(random_pick,song2);
                printf("Hint: Singer is Ed Sheeran\n");
                break;
            case 2:
                strcpy(random_pick,song3);
                printf("Hint: Singer is Dua Lipa\n");
                break;
        }

        do{
            printf("Comm|on Try to  Guess The song: ");
            scanf("%s",&guess);
```

```c
            if (strcmp(guess,random_pick)==0)
            {
                printf("Correct!!");
                break;
            }
            else
            {
                printf("Wrong Guess Try Again\n");
            }
        }
        while(strcmp(guess,random_pick)!=0 );
        return 0;
    }
```

Output –

```
E:\06Aug_Kevin_SE\C _ Assign

Hint: Singer is Ed Sheeran
Comm|on Try to  Guess The song: ShapeOfYou
Correct!!
--------------------------------
Process exited after 7.388 seconds with return value 0
Press any key to continue . . .
```

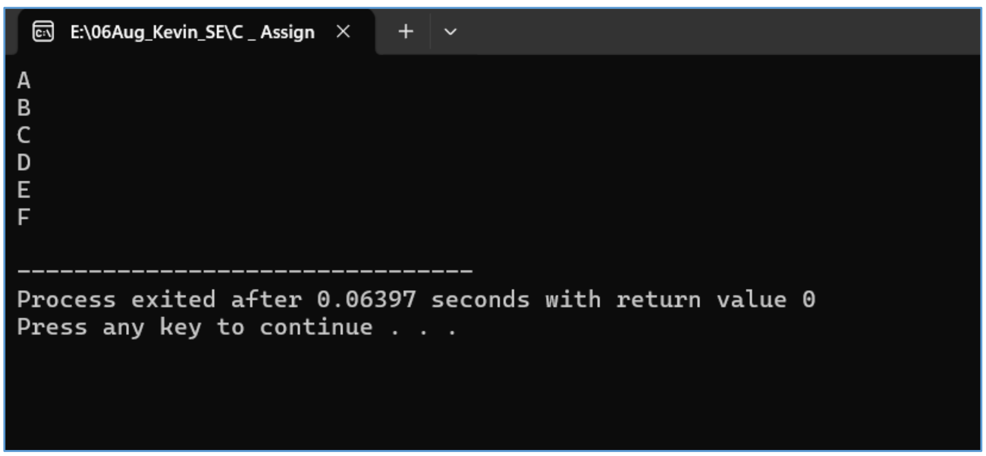
Que – 4

Explain with your own example the difference between entry-controlled and exit-controlled loops by writing a short code snippet for each (for/while vs do-while) and describing what happens if the loop condition is false at the start.

```
// Entry Control Loops are: 1) while 2)for
// Exit Control Loops are: 1)for
An entry-controlled loop checks the condition before running the code block,
while an exit-controlled loop runs the code block first before checking the
conditio
#include<stdio.h>
int main()
{
    char i='A';

    while (i<='Z')
    {
        printf("%c\n",i);
        i++;
    }
    printf("\n");

    return 0;
}
```

```
E:\06Aug_Kevin_SE\C _ Assign
A
B
C
D
E
F
G
H
I
J
K


--------------------------------
Process exited after 0.05194 seconds with return value 0
Press any key to continue . . .
```

```c
// Do - While Exit Control loop

#include<stdio.h>
int main()
{
    char i ='A';
    do {
        printf("%c\n",i);
        i++;
    }
    while (i<='F');
}
```



```c
// for Entry control loop

#include<stdio.h>
int main()
{
    int i,j;
    char ch;
    for(i=1;i<=5;i++)
    {
        ch = 'A';
        for(j=1;j<=i;j++)
        {
            printf("%c",ch);
            ch++;
        }
        printf("\n");
    }
}
```

```
E:\06Aug_Kevin_SE\C _ Assign
A
AB
ABC
ABCD
ABCDE

--------------------------------
Process exited after 0.01265 seconds with return value 0
Press any key to continue . . .
```

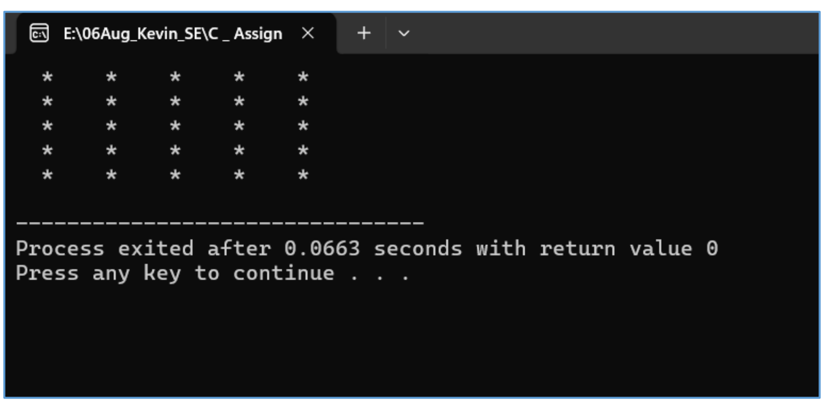
Session – 7

Que – 1

Use nested for loops to print a grid of emojis representing a 5x5 Instagram post feed, where each cell shows a 📷 symbol.

```
#include <stdio.h>

int main()
{
    for(int i=1;i<=5;i++)
    {
        for(int j =1;j<=5;j++)
        {
            printf("   *   ");
        }

        printf("\n");
    }
    return 0;
}
```

Output:

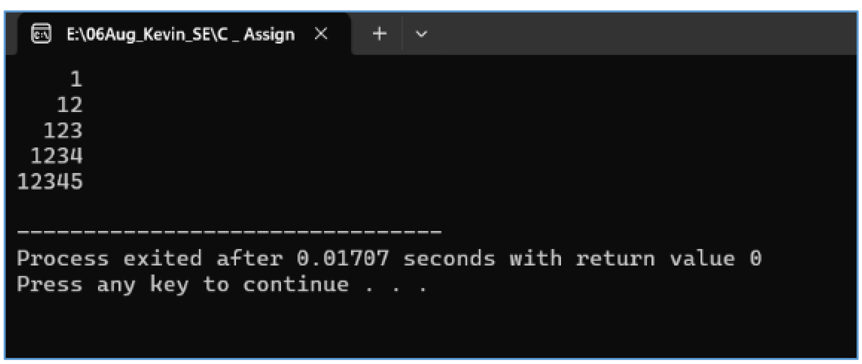


Que – 2

Build a right-angled triangle pattern using nested loops, where each row displays increasing numbers starting from 1, similar to how a leaderboard on a gaming app shows rank numbers.

```
#include<stdio.h>
int main()
{
    int i,j,k;
    int n=5;

    for (i=1;i<=n;i++)
    {
        for(j=n-1;j>=i;j--)
        {
                printf(" ");
        }
            for(k=1;k<=i;k++)
            {
                printf("%d",k);
            }
        printf("\n");
    }
    return 0;
}
```

Output:



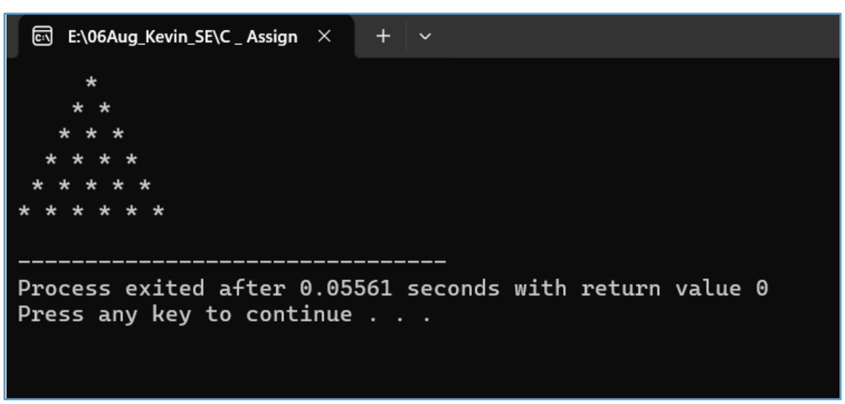
Que – 3

Create a pattern that prints a pyramid of stars (*) with 6 rows, centered like the loading animation you see on BookMyShow when a page is loading.

Hint: Use spaces to align the stars in the center for each row.

```
#include<stdio.h>
int main()
{
    int i,j,k;
    int n=6;

    for (i=1;i<=n;i++)
    {
        for(j=n-1;j>=i;j--)
        {
                printf(" ");
        }
            for(k=1;k<=i;k++)
            {
                printf("* ");
            }
        printf("\n");
    }
    return 0;
}
```

Output:

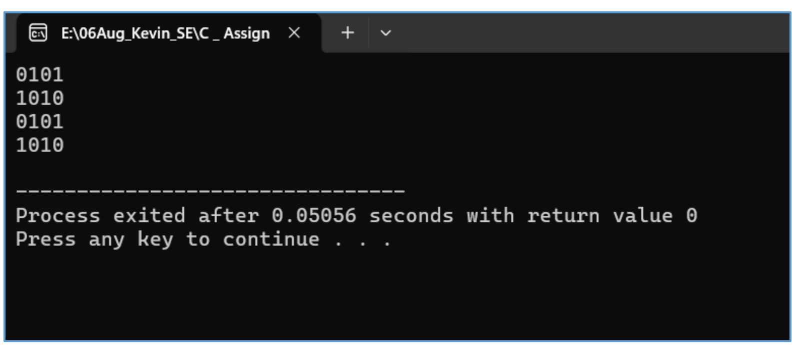


Que – 4

Write code using nested loops to print a pattern of alternating 0s and 1s in a grid, like the checkered background seen in some Spotify playlist covers (e.g., for a 4x4 grid, alternate 0 and 1 in each cell).

```c
#include <stdio.h>

int main()
{
    for(int i=1;i<=4;i++)
    {
        for(int j =1;j<=4;j++)
        {
            if ((i+j)%2 == 0)
            {
                printf("0");
            }
            else
            {
                printf("1");
            }
        }

        printf("\n");
    }
    return 0;
}
```

Output:



Que – 5

Modify your pyramid pattern code to accept the number of rows as user input, so the user can set the height of the pyramid before printing.

```c
#include <stdio.h>

int main()
{
    int size ;
    printf("Enter the Size of Pyramid--->");
    scanf("%d",&size);

    for(int i=1;i<=size;i++)
    {
        for(int j=1;j<=i;j++)
        {
            printf(" ^ ");
        }

        printf("\n");
    }
    return 0;
}
```

Output:

```
E:\06Aug_Kevin_SE\C _ Assign
Enter the Size of Pyramid--->10
 ^
 ^  ^
 ^  ^  ^
 ^  ^  ^  ^
 ^  ^  ^  ^  ^
 ^  ^  ^  ^  ^  ^
 ^  ^  ^  ^  ^  ^  ^
 ^  ^  ^  ^  ^  ^  ^  ^
 ^  ^  ^  ^  ^  ^  ^  ^  ^
 ^  ^  ^  ^  ^  ^  ^  ^  ^  ^

--------------------------------
Process exited after 7.899 seconds with return value 0
Press any key to continue . . .
```